

PMA Seminar Notes

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About These Notes

These notes are intended to accompany the reading of Walter Rudin's *Principles of Mathematical Analysis*, rather than to serve as a self-contained replacement for the original text.

The statements of definitions, theorems, and exercises are generally reproduced from Rudin. Proofs and solutions are expanded, reorganized, or supplemented when additional detail seems useful. When Rudin's proof requires no further elaboration, it is omitted and only the statement is recorded.

Remarks, examples, and auxiliary results not appearing in the original text are occasionally added for clarification or later reference. Such additions are indicated when appropriate.

Contents

About These Notes	ii
1 The Real and Complex Number Systems	1
Exercises	1
2 Basic Topology	5
Exercises	5
3 Numerical Sequences and Series	8
Exercises	8
4 Continuity	10
Exercises	10

1 The Real and Complex Number Systems

Exercises

Exercise 1. Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that

$$\inf A = -\sup(-A).$$

Proof. Since A is bounded below, the infimum of A exists in $\mathbb{R}$. If $a \in A$ then $\inf A \leq a$. Hence,

$$-a \leq -\inf A \text{ for all } a \in A, \quad (1)$$

and it follows that $-A$ is bounded above. Thus, $\sup(-A)$ exists in $\mathbb{R}$. Since $-\inf A$ is an upper bound of $-A$ from (1), we have $\sup(-A) \leq -\inf A$, hence

$$\inf A \leq -\sup(-A). \quad (2)$$

Since $-a \leq \sup(-A)$ for every $a \in A$, we obtain $-\sup(-A) \leq a$ for all $a \in A$. Therefore, $-\sup(-A)$ is a lower bound of A and

$$-\sup(-A) \leq \inf A. \quad (3)$$

Combining (2) and (3), we conclude that $\inf A = -\sup(-A)$. $\square$

Exercise 2. Fix $b > 1$.

(a) If m, n, p, q are integers, $n > 0, q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

(b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.

(c) If x is real, define $B(x)$ to be the set of all numbers b^t , where t is rational and $t \leq x$. Prove that

$$b^r = \sup B(r)$$

when r is rational. Hence it makes sense to define

$$b^x = \sup B(x)$$

for every real x .

(d) Prove that $b^{x+y} = b^x b^y$ for all real x and y .

Proof.

(a) Let $b^{mq} = b^{np} = x$. Since n and q are positive integers and $b > 0$, by Theorem 1.21, we have

$$b^m = x^{1/q} \text{ and } b^p = x^{1/n}.$$

It follows that

$$(b^m)^{1/n} = (x^{1/q})^{1/n} = (x^{1/n})^{1/q} = (b^p)^{1/q},$$

since $[(x^{1/q})^{1/n}]^{nq} = [(x^{1/n})^{1/q}]^{nq} = x$.

- (b) Let m, n, p, q be integers and $n > 0, q > 0$ and let $r = m/n$ and $s = p/q$. Then by the Corollary to Theorem 1.21, we obtain

$$\begin{aligned}
 b^{r+s} &= b^{\frac{m}{n} + \frac{p}{q}} = b^{\frac{mq+np}{nq}} \\
 &= (b^{mq+np})^{\frac{1}{nq}} \\
 &= (b^{mq}b^{np})^{\frac{1}{nq}} \\
 &= (b^{mq})^{\frac{1}{nq}} (b^{np})^{\frac{1}{nq}} \\
 &= b^{\frac{mq}{nq}} b^{\frac{np}{nq}} = b^{\frac{m}{n}} b^{\frac{p}{q}} = b^r b^s.
 \end{aligned}$$

- (c) Note that by Theorem 1.21, $b^r > 0$ if $b > 1$ and r is rational.

Let n be a positive integer and let $0 < a < 1$ be real. We first show that $0 < a^{1/n} < 1$.

Suppose to the contrary that $a^{1/n} \geq 1$. Then we obtain

$$1 \leq (a^{1/n})^n = a < 1,$$

a contradiction. Therefore, if $0 < a < 1$ then $0 < a^{1/n} < 1$, where n is a positive integer.

Suppose r and s are rational. Let m, n, p , and q be integers and let n and q be positive. Then we write $r = m/n, s = p/q$.

Suppose $r < s$. Then

$$r - s = \frac{m}{n} - \frac{p}{q} = \frac{mq - np}{nq} < 0,$$

hence

$$0 < b^{mq-np} = \frac{1}{b^{np-mq}} < 1$$

and

$$0 < b^{r-s} = b^{\frac{mq-np}{nq}} = (b^{mq-np})^{1/nq} < 1.$$

Thus, if $r < s$ then

$$b^r = b^{(r-s)+s} = b^{r-s} b^s < b^s.$$

If t is rational and $t \leq r$ then $b^t \leq b^r$. It follows that $B(r)$ is bounded above by b^r and $\sup B(r)$ exists in $\mathbb{R}$. Since b^r is an upper bound of $B(r)$,

$$\sup B(r) \leq b^r.$$

Since $b^r \in B(r)$,

$$b^r \leq \sup B(r).$$

We conclude that $b^r = \sup B(r)$.

- (d) Let t_1 and t_2 be rational. If $t_1 \leq x$ and $t_2 \leq y$ then $t_1 + t_2 \leq x + y$. Thus, we have

$$b^{t_1} b^{t_2} = b^{t_1+t_2} \leq \sup B(x+y) = b^{x+y}.$$

Fix $t_1 \leq x$. Then

$$b^{t_2} \leq b^{x+y}/b^{t_1} \text{ for all } t_2 \leq y.$$

Hence,

$$b^y = \sup B(y) \leq b^{x+y}/b^{t_1}.$$

Since the choice of t_1 was arbitrary,

$$b^{t_1} \leq b^{x+y}/b^y \text{ for all } t_1 \leq x,$$

and it follows that

$$b^x = \sup B(x) \leq b^{x+y}/b^y.$$

Therefore, we obtain

$$b^x b^y \leq b^{x+y}. \quad (4)$$

Notice that

$$0 < b^{t_1} \leq b^x \text{ for every } t_1 \leq x$$

and

$$0 < b^{t_2} \leq b^y \text{ for every } t_2 \leq y.$$

Hence, for every $t_1 \leq x$ and $t_2 \leq y$,

$$b^{t_1+t_2} = b^{t_1} b^{t_2} \leq b^x b^y.$$

Now, to obtain

$$b^{x+y} = \sup B(x+y) \leq b^x b^y \quad (5)$$

from above inequality, we need the following statement:

$$b^x = \sup \{b^t \mid t \in \mathbb{Q}, t < x\}.$$

Let us prove it. Notice that

$$b^n - 1 = (b - 1)(b^{n-1} + \cdots + 1) \geq n(b - 1)$$

for any positive integer n . Put $c = b^{1/n} > 1$. Then we have

$$b - 1 \geq n(b^{1/n} - 1).$$

Hence, if $t > 1$ and $n > (b - 1)/(t - 1)$ then

$$\frac{n(b^{1/n} - 1)}{t - 1} \leq \frac{b - 1}{t - 1} < n,$$

and it follows that

$$b^{1/n} < t.$$

Note that if $t > 1$ then $b^{1/n} < t$ for sufficiently large n .

Let $B_0(x) = \{b^t \mid t \in \mathbb{Q}, t < x\}$. We need to show that b^x is the supremum of $B_0(x)$. It is clear that b^x is an upper bound of $B_0(x)$. Thus, it remains to prove that if $y < b^x$ then y is not an upper bound.

Suppose $0 < y < b^x$. Then $b^x/y > 1$ and it follows that

$$b^{1/n} < b^x/y$$

for sufficiently large n . Hence, we have

$$y < \frac{b^x}{b^{1/n}} < b^x.$$

By the density of rationals, choose a rational t such that $x - (1/n) < t < x$. Since $x < s < t + (1/n)$ for some rational s ,

$$b^x \leq b^s < b^{t+(1/n)} = b^t b^{1/n}$$

and

$$y < \frac{b^x}{b^{1/n}} < b^t.$$

Thus, y is not an upper bound of $B_0(x)$ and therefore $b^x = \sup B_0(x)$.

Now, let t be a rational and let $t < x + y$. Pick $t_2 \in \mathbb{Q}$ such that $t - x < t_2 < y$ and set $t_1 = t - t_2$. Then for every $t \in \mathbb{Q}$ with $t < x + y$, there are rational $t_1 < x$ and $t_2 < y$ such that $t = t_1 + t_2 < x + y$, hence $b^t = b^{t_1+t_2} = b^{t_1} b^{t_2} \leq b^x b^y$. Thus, we obtain (5).

Combining (4) and (5), we conclude that

$$b^{x+y} = b^x b^y.$$

□

Exercise 3. Prove that no order can be defined in the complex field that turns it into an ordered field.

Proof. Suppose to the contrary that there exists an order $<$ on $\mathbb{C}$ which turns $\mathbb{C}$ into an ordered field. Let $\mathbf{0}$ and $\mathbf{1}$ denote the additive and multiplicative identities of the complex field, respectively. Then $\mathbf{0} = (0, 0)$ and $\mathbf{1} = (1, 0)$.

Note that $i = (0, 1) \neq \mathbf{0}$. If $x \neq \mathbf{0}$ then $x^2 > \mathbf{0}$. It follows that

$$i^2 = (0, 1)(0, 1) = (-1, 0) > \mathbf{0}.$$

Since $(-1, 0) + (1, 0) = \mathbf{0}$, $(-1, 0)$ is the additive inverse of $(1, 0)$. Since $\mathbf{1} > \mathbf{0}$ and if $x > \mathbf{0}$ then $-x < \mathbf{0}$ in any ordered field, we obtain

$$(-1, 0) = -(1, 0) = -\mathbf{1} < \mathbf{0}.$$

This contradicts the Trichotomy Property.

□

2 Basic Topology

Exercises

Exercise 4. Let E' be the set of all limit points of a set E . Prove that E' is closed. Prove that E and $\overline{E}$ have the same limit points. (Recall that $\overline{E} = E \cup E'$.) Do E and E' always have the same limit points?

Proof. Let p be a limit point of E' and let N be any neighborhood of p . Then N contains a point $q \neq p$ such that $q \in E'$. Since N is open, there exists a neighborhood N_q of q such that $N_q \subset N$. Without loss of generality, assume that N_q does not contain p . For if $p \in N_q$ then we take $r = d(p, q) > 0$, hence $p \notin N_r(q) \subset N_q \subset N$.

Recall that q is a limit point of E ($q \in E'$), we see that N_q contains a point $s \neq q$ such that $s \in E$. Since $p \notin N_q$, we have $p \neq s$. Therefore, $N \supset N_q$ contains a point $s \neq p$ such that $s \in E$. Since N was arbitrary, p is a limit point of E and $p \in E'$.

Since the choice of p was arbitrary, every limit point of E' is a point of E' . Thus, E' is closed.

Next, let us prove that E and $\overline{E}$ have the same limit points. Suppose p is a limit point of E . Then every neighborhood of p contains a point $q \neq p$ such that $q \in E \subset E \cup E' = \overline{E}$. Thus p is a limit point of $\overline{E}$.

Conversely, suppose p is a limit point of $\overline{E}$. Then every neighborhood N of p contains a point $q \neq p$ such that $q \in \overline{E}$. Note that $q \in E$ or $q \in E'$.

If $q \in E$ then we are done. Otherwise, suppose $q \in E'$. Since N is open, there is a neighborhood N_q of q such that $N_q \subset N$. Without loss of generality, assume N_q does not contain p . Since q is a limit point of E , N_q contains a point $s \neq q$ such that $s \in E$. Hence, N contains a point $s \neq p$ (since $p \notin N_q$) such that $s \in E$. Since N was arbitrary, p is a limit point of E .

We conclude that E and $\overline{E}$ have the same limit points.

Finally, let $E = \{\frac{1}{n} \mid n \in \mathbb{N}\}$. It is clear that $E' = \{0\}$ but there is no limit point of E' . Hence it is possible that E and E' have different limit points. $\square$

Exercise 5. Let $A_1, A_2, A_3, \dots$ be subsets of a metric space.

(a) If $B_n = \bigcup_{i=1}^n A_i$, prove that $\overline{B_n} = \bigcup_{i=1}^n \overline{A_i}$, for $n = 1, 2, 3, \dots$

(b) If $B = \bigcup_{i=1}^{\infty} A_i$, prove that $\overline{B} \supset \bigcup_{i=1}^{\infty} \overline{A_i}$.

Show, by an example, that this inclusion can be proper.

Proof. To prove (a), by mathematical induction, it suffices to prove that $\overline{A \cup B} = \overline{A} \cup \overline{B}$. We need to prove that $(A \cup B)' = A' \cup B'$.

Suppose $p \in (A \cup B)'$. Assume, for contradiction, that p is neither a limit point of A nor B . Then there are neighborhoods N_1 and N_2 of p such that $N_1 \cap (A \setminus \{p\})$ and $N_2 \cap (B \setminus \{p\})$ are empty, respectively.

Let $N = N_1 \cap N_2$. Then we have

$$\begin{aligned} [N \cap (A \setminus \{p\})] \cup [N \cap (B \setminus \{p\})] &= N \cap [(A \setminus \{p\}) \cup (B \setminus \{p\})] \\ &= N \cap [(A \cup B) \setminus \{p\}] = \emptyset. \end{aligned}$$

This contradicts our assumption. Therefore, $(A \cup B)' \subset A' \cup B'$.

Conversely, suppose $p \in A' \cup B'$. Then $p \in A'$ or $p \in B'$. If $p \in A'$, then every neighborhood of p contains a point $q \neq p$ such that $q \in A \subset A \cup B$, hence $p \in (A \cup B)'$. Similarly, if $p \in B'$ then p is a limit point of $A \cup B$, hence $p \in (A \cup B)'$. This proves $A' \cup B' \subset (A \cup B)'$.

We have shown that $(A \cup B)' \subset A' \cup B'$ and $A' \cup B' \subset (A \cup B)'$. Hence $(A \cup B)' = A' \cup B'$ and it follows that

$$\begin{aligned}\overline{A \cup B} &= (A \cup B) \cup (A \cup B)' = (A \cup B) \cup (A' \cup B') \\ &= (A \cup A') \cup (B \cup B') = \overline{A} \cup \overline{B}.\end{aligned}$$

This is the desired result.

Next, let us prove (b). Suppose $p \in \bigcup_{i=1}^{\infty} \overline{A_i}$. Then $p \in \overline{A_i}$ for some positive integer i . If $p \in A_i$ then $p \in B$. If $p \in A'_i$ then every neighborhood of p contains a point $q \neq p$ such that $q \in A_i \subset \bigcup_{i=1}^{\infty} A_i = B$. Therefore, p is a limit point of B and $p \in B'$.

Hence, if $p \in \bigcup_{i=1}^{\infty} \overline{A_i}$ then $p \in B$ or $p \in B'$. This proves $\bigcup_{i=1}^{\infty} \overline{A_i} \subset \overline{B}$.

Finally, we show that this inclusion can be proper. For each $n \in \mathbb{N}$, define

$$A_n = \left\{ x \in \mathbb{R} \mid \frac{1}{n} < x \right\}.$$

Then we see that B is the set of positive real numbers. Observe that 0 is not a limit point of A_n and $0 \notin A_n$ for $n = 1, 2, 3, \dots$. It follows that $0 \notin \bigcup_{i=1}^{\infty} \overline{A_i}$. But it is clear that $0 \in \overline{B}$. $\square$

Exercise 6. Is every point of every open set $E \subset \mathbb{R}^2$ a limit point of E ? Answer the same question for closed sets in $\mathbb{R}^2$.

Proof. Suppose $p \in E$ is not a limit point of E . Then there exists a neighborhood N' of p such that $N' \cap E = \{p\}$. Since E is open, choose a neighborhood N'' of p such that $N'' \subset E$. Then we have

$$N = (N'' \cap N') \subset (E \cap N') = \{p\}.$$

Since $p \in N$, it follows that $N = \{p\}$. Since N is an intersection of two neighborhoods of the same point p in $\mathbb{R}^2$, N is a neighborhood of p . But in $\mathbb{R}^2$, every neighborhood is infinite. This is a contradiction.

Next, let $F = \{\mathbf{x}\} \subset \mathbb{R}^2$. It is clear that F is a closed subset of $\mathbb{R}^2$ but $\mathbf{x} \in F$ is not a limit point of F . $\square$

Exercise 7. Let $K \subset \mathbb{R}^1$ consist of 0 and the numbers $1/n$, for $n = 1, 2, 3, \dots$. Prove that K is compact directly from the definition (without using the Heine-Borel theorem).

Proof. Let $\{G_\alpha\}$ be an open cover of K . Since $0 \in K$, there exists an α_0 such that $0 \in G_{\alpha_0}$. Since every G_α is open, choose a neighborhood N of 0 with radius r such that $N \subset G_{\alpha_0}$.

If $r > 1$ then $K \subset G_{\alpha_0}$ hence we are done.

If $0 < r \leq 1$ then the set

$$E = \{1/n \mid n \in \mathbb{N}, r \leq 1/n\}$$

is finite by the Archimedean Property. Let $x_1, \dots, x_k$ be the points of E . Since $x_i \in K \subset \bigcup_\alpha G_\alpha$ ($1 \leq i \leq k$), there exists α_i such that $x_i \in G_{\alpha_i}$ for $i = 1, \dots, k$. Thus, we have finitely many indices $\alpha_0, \alpha_1, \dots, \alpha_k$ such that

$$K \subset G_{\alpha_0} \cup G_{\alpha_1} \cup \dots \cup G_{\alpha_k}.$$

We conclude that K is compact. $\square$

Exercise 8. Give an example of an open cover of the segment $(0, 1)$ which has no finite subcover.

Proof. For each $n \in \mathbb{N}$, let

$$G_n = \left\{ x \in \mathbb{R} \mid 0 < x < 1 - \frac{1}{n} \right\}.$$

If $0 < x < 1$, then there exists a positive integer n such that $1 < n(1 - x)$, hence $0 < x < 1 - \frac{1}{n}$ so that $x \in G_n$. It follows that the collection $\{G_n\}_{n \in \mathbb{N}}$ is an open cover of $(0, 1)$. But it is clear that any finite subcollection of $\{G_n\}_{n \in \mathbb{N}}$ cannot cover $(0, 1)$. $\square$

Exercise 9.

- (a) If A and B are disjoint closed sets in some metric space X , prove that they are separated.
- (b) Prove the same for disjoint open sets.
- (c) Fix $p \in X$, $\delta > 0$, define A to be the set of all $q \in X$ for which $d(p, q) < \delta$, define B similarly, with $>$ in place of $<$. Prove that A and B are separated.
- (d) Prove that every connected metric space with at least two points is uncountable.

Hint: use(c).

Proof.

- (a) Suppose A and B are disjoint closed subsets of a metric space X . Since $A = \overline{A}$ and $B = \overline{B}$ by the Theorem 2.27, we have $\overline{A} \cap B = A \cap \overline{B} = A \cap B = \emptyset$. This proves (a).
- (b) Let A and B be disjoint open subsets of X . Since $A \cap B = \emptyset$, we have $A \subset B^c$. Since B^c is closed (notice B is open), the Theorem 2.27 implies that $\overline{A} \subset B^c$, hence $\overline{A} \cap B = \emptyset$. By symmetry, the same argument proves that $A \cap \overline{B} = \emptyset$. Therefore, A and B are separated.
- (c) Fix $p \in X$ and $\delta > 0$. Then by the definition of A , it is a neighborhood in X , hence it is open.

We wish to prove that B is open. Let $q \in B$ be given and put $r = d(p, q) - \delta > 0$. If $x \in X$ and $d(q, x) < r$ then

$$\begin{aligned} d(p, x) &\geq d(p, q) - d(q, x) \\ &= r + \delta - d(q, x) > \delta, \end{aligned}$$

hence $x \in B$. It follows that $N_r(q) \subset B$ and therefore B is open.

Since A and B are disjoint open sets in X , (b) shows that they are separated.

- (d) Let X be a connected metric space. Suppose p and q are two distinct points in X and put $r = d(p, q) > 0$. For every $\delta \in (0, r)$, define

$$A_\delta = \{x \in X \mid d(p, x) < \delta\}, \quad B_\delta = \{x \in X \mid d(p, x) > \delta\}.$$

It follows from (c) that A_δ and B_δ are separated. Thus it is impossible that $X = A_\delta \cup B_\delta$ and hence there exists a point $x_\delta \in X$ such that $d(p, x_\delta) = \delta$ for each $\delta \in (0, r)$.

Let $f(\delta) = x_\delta$ for $0 < \delta < r$. If $\delta_1 \neq \delta_2$ then $f(\delta_1) \neq f(\delta_2)$, hence f is a 1-1 function from $(0, r)$ to X . Since $(0, r)$ is an uncountable subset of $\mathbb{R}$, we conclude that X is uncountable.

□

3 Numerical Sequences and Series

Exercises

Exercise 10. Suppose $a_n > 0$, $s_n = a_1 + \cdots + a_n$, and $\sum a_n$ diverges.

(a) Prove that $\sum \frac{a_n}{1+a_n}$ diverges.

(b) Prove that

$$\frac{a_{N+1}}{s_{N+1}} + \cdots + \frac{a_{N+k}}{s_{N+k}} \geq 1 - \frac{s_N}{s_{N+k}}$$

and deduce that $\sum \frac{a_n}{s_n}$ diverges.

(c) Prove that

$$\frac{a_n}{s_n^2} \leq \frac{1}{s_{n-1}} - \frac{1}{s_n}$$

and deduce that $\sum \frac{a_n}{s_n^2}$ converges.

(d) What can be said about

$$\sum \frac{a_n}{1+na_n} \text{ and } \sum \frac{a_n}{1+n^2a_n}?$$

Proof. (a) Assume a_n diverges. If $\{n_i\}$ is a sequence of positive integers with $n_1 < n_2 < n_3 < \cdots$, then we have

$$\sum_{i=1}^{n_k} \frac{a_i}{1+a_i} \geq \sum_{i=1}^k \frac{a_{n_i}}{1+a_{n_i}}.$$

Since a_n diverges, we have $0 < s = \limsup_{n \rightarrow \infty} a_n \leq \infty$. Pick a subsequence $\{a_{n_i}\}_{i \in \mathbb{N}}$ such that $a_{n_i} \rightarrow s$ and let $k \rightarrow \infty$. Then right side of the last inequality diverges, since $\frac{a_{n_i}}{1+a_{n_i}}$ does not converge to 0. Hence left hand side diverges.

Next, assume a_n converges and let $a_n \rightarrow 0$. (Otherwise, $\sum \frac{a_n}{1+a_n}$ diverges.) Then there exists an integer N_1 such that $a_n < 1$ for $n \geq N_1$.

Suppose, for a contradiction, that $\sum \frac{a_n}{1+a_n}$ converges. Let $\varepsilon > 0$ be given. Choose an integer N_2 such that

$$m \geq n \geq N_2 \text{ implies } \sum_{k=n}^m \frac{a_k}{1+a_k} \leq \varepsilon/2.$$

Now, let $N = \max\{N_1, N_2\}$ and let $\alpha = \sup_{k \geq N} a_k$. If $m \geq n \geq N$ then

$$\begin{aligned} \frac{a_n}{1+a_n} + \cdots + \frac{a_m}{1+a_m} &\geq \frac{1}{1+\alpha}(a_n + \cdots + a_m) \\ &= \frac{1}{1+\alpha} \sum_{k=n}^m a_k, \end{aligned}$$

hence

$$\sum_{k=n}^m a_k \leq \frac{1+\alpha}{2} \varepsilon \leq \varepsilon.$$

This contradicts our hypothesis that $\sum a_n$ diverges.

(b) Since $a_n > 0$, we have

$$\begin{aligned}\frac{a_{N+1}}{s_{N+1}} + \cdots + \frac{a_{N+k}}{s_{N+k}} &\geq \frac{a_{N+1} + \cdots + a_{N+k}}{s_{N+k}} \\ &= \frac{s_{N+k} - s_N}{s_{N+k}} \\ &= 1 - \frac{s_N}{s_{N+k}}.\end{aligned}$$

for every positive integer N . This yields the desired inequality.

Suppose $\sum \frac{a_n}{s_n}$ converges. Choose an integer N such that $m \geq n \geq N$ implies

$$\sum_{k=n}^m \frac{a_k}{s_k} \leq 1/2.$$

Hence we have

$$1 - \frac{s_N}{s_{N+k}} \leq \frac{a_{N+1}}{s_{N+1}} + \cdots + \frac{a_{N+k}}{s_{N+k}}$$

for fixed N and for all $k \in \mathbb{N}$. Letting $k \rightarrow \infty$, we obtain $1 \leq \frac{1}{2}$, a contradiction.

(c) Observe that

$$\begin{aligned}\frac{a_n}{s_n^2} &\leq \frac{a_n}{s_{n-1}s_n} \\ &= \frac{a_n}{s_n - s_{n-1}} \left(\frac{1}{s_{n-1}} - \frac{1}{s_n} \right) \\ &= \frac{1}{s_{n-1}} - \frac{1}{s_n}.\end{aligned}$$

Taking summation from $n = 2$ of both sides and letting $n \rightarrow \infty$, by telescoping, we conclude that $\sum \frac{a_n}{s_n^2}$ converges.

(d) We first prove that the latter one converges. Observe that

$$\frac{a_n}{1 + n^2 a_n} \leq \frac{a_n}{n^2 a_n} = \frac{1}{n^2},$$

which implies that $\sum \frac{a_n}{1 + n^2 a_n}$ converges.

However, there is nothing that we can say about the former one. For instance, if we put $a_n = \frac{1}{n}$ then $\sum \frac{a_n}{1 + n a_n} = \sum \frac{1}{2n}$ diverges. But if we take $a_{m^2} = 1/m$ for $m = 1, 2, 3, \dots$, while $a_n = 2^{-n}$ if n is not a square, then the sum converges.¹

□

Exercise 11. suppose $\{p_n\}$ is a Cauchy sequence in a metric space X , and some subsequence $\{p_{n_i}\}$ converges to a point $p \in X$. Prove that the full sequence $\{p_n\}$ converges to p .

Proof. Fix $\varepsilon > 0$. Since $\{p_n\}$ is Cauchy, there exists an integer N such that $d(p_n, p_m) < \varepsilon/2$ if $n \geq N$ and $m \geq N$. Since a subsequence $\{p_{n_i}\}$ converges to p , choose an integer $k \geq N$ such that $d(p_{n_k}, p) < \varepsilon/2$. Hence, $n \geq N$ implies

$$\begin{aligned}d(p_n, p) &\leq d(p_n, p_{n_k}) + d(p_{n_k}, p) \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon.\end{aligned}$$

This is the desired result.

□

¹The last example is from this: <https://math.stackexchange.com/questions/237691/baby-rudin-chapter-3-exercise-11d>

4 Continuity

Exercises

Exercise 12. Let f be a continuous real function on a metric space X . Let $Z(f)$ (the *zero set* of f) be the set of all $p \in X$ at which $f(p) = 0$. Prove that $Z(f)$ is closed.

Proof. Let W be a complement of $Z(f)$ in X . We prove that W is open.

Suppose $p \in W$. Then we have $f(p) \neq 0$, or equivalently, $|f(p)| > 0$. Let V be a neighborhood of $f(p)$ whose radius equals $\frac{|f(p)|}{2}$. Since V is open in $\mathbb{R}$ and f is continuous, $f^{-1}(V)$ is also open in X .

Now, choose a neighborhood N of p such that $N \subset f^{-1}(V)$. Since $f^{-1}(V)$ is a subset of W , it follows that p is an interior point of W . Since $p \in W$ was arbitrary, this proves that W is open. We conclude that $Z(f)$ is closed in X . $\square$

Exercise 13. Let f and g be continuous mappings of a metric space X into a metric space Y , and let E be a dense subset of X . Prove that $f(E)$ is dense in $f(X)$. If $g(p) = f(p)$ for all $p \in E$, prove that $g(p) = f(p)$ for all $p \in X$. (In other words, a continuous mapping is determined by its values on a dense subset of its domain.)

Proof. We first prove that $f(E)$ is dense in $f(X)$. Let $q \in f(X)$ be arbitrary. We need to prove that q is a point of $f(E)$, or a limit point of $f(E)$.

If $q \in f(E)$ then we are done.

Suppose $q \notin f(E)$ and pick a point $p \in X$ such that $f(p) = q$. Since E is dense in X , we have $p \in E$ or $p \in E'$. Since $p \notin E$ (for otherwise, $q = f(p) \in f(E)$), we see that p is a limit point of E .

By continuity of f , we obtain $f(x) \rightarrow f(p)$ as $x \rightarrow p$. Now, choose a sequence $\{x_n\}$ in E such that $x_n \neq p$ and $x_n \rightarrow p$. Then $f(x_n) \rightarrow f(p)$. Hence, for every $\varepsilon > 0$, there exists a point $x_n \in E$ such that

$$d_Y(f(x_n), f(p)) < \varepsilon.$$

Since $f(p) \notin f(E)$, it follows that $f(x_n) \neq f(p)$ for each positive integer n . Therefore, $q = f(p)$ is a limit point of $f(E)$.

We conclude that $f(E)$ is dense in $f(X)$.

Next, suppose $g(p) = f(p)$ for all $p \in E$, where f and g are continuous mappings of X into Y . We need to prove that $f(p) = g(p)$ for $p \in X$.

Let $p \in X$. Since E is dense in X , $p \in E$ or $p \in E'$ for every point $p \in X$.

If $p \in E$ then there is nothing to prove.

If $p \in E'$ then $f(x) \rightarrow f(p)$ and $g(x) \rightarrow g(p)$ as $x \rightarrow p$, since f and g are continuous. Now choose a sequence $\{y_n\}$ in E such that $y_n \neq p$ and $y_n \rightarrow p$. Since $y_n \in E$ for every positive integer n , we have $g(y_n) = f(y_n)$ by our hypothesis. Letting $n \rightarrow \infty$, we obtain $g(p) = f(p)$.

This completes the proof. $\square$